THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE MEET #3

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge Meet #3 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the expression `0b100100`?

A. 37_{10} B. 203_4 C. 44_8 D. 26_{16} E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. `2`
B. `17`
C. `32`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
out.println(5 * 6+8 / 4 % 5);
```

QUESTION 3

What is output by the code to the right ($'s represent spaces)?

A. `$$$$$$$$$$$Ch`
B. `$$Chick$$$$$$$`
C. `$$$Chicken`
D. `$$$$$$$$$$$$Ch`
E. `$$$$$$$$$$$Chi`

```
out.printf("%12.2s", "Chicken");
```

QUESTION 4

What is output by the code to the right?

A. `mpsci.com`
B. `mpsci.com219`
C. `com219`
D. `mpsci`
E. `apluscomp219`

```
String s="apluscompsci.com";
s+=s.charAt(2) + s.charAt(6);
s=s.substring(7);
out.println(s);
```

QUESTION 5

What is output by the code to the right?

A. `True`
B. `False`
C. `false`
D. `true`
E. There is no output due to a syntax error.

```
int b=5;
boolean a = b>0 && b <10;
boolean c =!(a > 9)^b && !b^a < 2;
out.println(c);
```

QUESTION 6

What is output by the code to the right?

A. `36.0`
B. `36`
C. `37.0`
D. `37`
E. There is no output due to an error.

```
double g = Math.pow(6, 2) + .007;
out.println(Math.ceil(g));
```

QUESTION 7

What is the output by the code to the right?

A. `Yoted`
B. `yeet`
C. `Yoteyeet`
D. `Yotedyeet`
E. `yat`

```
int y="chicken".compareTo("tenders");
if(y<0)
  out.print("Yoted");
if(y>0)
  out.print("yat");
else
  out.print("yeet");
```

QUESTION 8

What is the output by the code to the right?

A. `223`
B. `789,202,23`
C. `78,920,223`
D. `789`
E. `78920`

```
out.printf("%,3d", 78920223);
```

QUESTION 9

What is the sum of the ASCII values in `String s` to the right (including spaces)?

A. `902`
B. `789`
C. `725`
D. `757`
E. There is no output due to a syntax error.

```
String s = " C H I c K E N";
```

QUESTION 10

What is the output by the code to the right ?

A. `Ranch`
B. `yeeet`
C. `null`
D. `Ranchyeet`
E. There is no output due to an error.

```
String [] joe = new String[25];
joe[4] = "Chicken";
joe[6] = "Tenders";
joe[8] = "With";
joe[10] = "Ranch";
joe[24] = "yeeet";
out.println(joe[22]);
```

QUESTION 11

What is the output by the code to the right?

A. `9`
B. `17`
C. `-1`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
String g = " chicken fingers
chicken";
Scanner file = new Scanner(g);
String h = file.next();
g = file.next();
int d =
g.indexOf(file.findInLine(h));
out.println(d);
```

QUESTION 12

What is the output by the code to the right ?

A. `67`
B. `65`
C. `89`
D. `24`
E. `0`

```
int y = 89;
int t = 65;
System.out.println(t^y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. `III, IV, II, I`
B. `IV, III, II, I`
C. `I, IV, III, II`
D. `I, III, II, IV`
E. `I, II, IV, III`

```
I. ()
II. +
III. /=
IV. &(bitwise)
```

QUESTION 14

What is the output by the code to the right ?

A. 4
B. 16
C. 8
D. 64
E. 128

```
out.println(Short.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. 95
B. 89
C. 88
D. 2
E. There is no output due to a syntax error.

```
ArrayList<Integer> a;
a = new ArrayList <Integer>();
a.add(95);
a.add(89);
a.add(88);
System.out.println(a.set(2,2));
```

QUESTION 16

What is the output by the code to the right ?

A. 161
B. 254
C. 253
D. 245
E. There is no output due to a syntax error.

```
out.println(212|89^44);
```

QUESTION 17

What is the output by the code to the right ?

A. LfgrF
B. LrgfF
C. L[f, g, r]L
D. FfgrL
E. There is no output due to an error.

```
char [] joey = new char [3];
joey[0] = 'f';joey[1] = 'g';
joey[2] = 'r';
StringBuffer s=new StringBuffer();
s.append('F');
s.append(joey);
s.append('L');
s.reverse();
out.println(s);
```

QUESTION 18

What could replace **<1*>** in the code to the right, so that the code will work as intended?

A. <String,String>
B. <char, char[]>
C. <char,String>
D. <Character,String>
E. More than one of the choices above is correct.

QUESTION 19

Assuming **<1*>** was filled correctly, what is output by the code to the right ?

A. domino
B. null
C. onimod
D. d
E. There is no output due to a runtime error.

```
TreeMap<1*>tm;
tm=new TreeMap<1*>();
tm.put('a', "apple");
tm.put('b', "banana");
tm.put('c', "crackers");
tm.put('d', "domino");
tm.put('e', "eggplant");
out.println(tm.get("d"));
```

QUESTION 20

What is the value of 1011001_2?

A. 87_{10}
B. 324_5
C. 58_{16}
D. 74_{12}
E. None of the above.

QUESTION 21

What is the output by the code to the right ?

A. `[39, 57, 898, 212]`

B. `[212, 898, 57, 39]`

C. `[39, 898, 212, 57]`

D. `[57, 898, 212, 39]`

E. There is no output due to a runtime error.

```
ArrayList<Integer> a;
a = new ArrayList<Integer>();
a.add(212);a.add(898);
a.add(57);a.add(39);
Collections.rotate(a, 3);
Collections.sort(a);
Collections.reverse(a);
Collections.rotate(a, 1);
out.println(a);
```

QUESTION 22

Which of the following concepts is demonstrated by the classes to the right ?

A. abstraction

B. downcasting

C. encapsulation

D. inheritance

E. More than one of the above.

QUESTION 23

What is the output by the line marked `//1` ?

A. `4`

B. `2`

C. `8`

D. `16`

E. There is no output due to a syntax error.

QUESTION 24

What is the output by the line marked `//2` ?

A. `6`

B. `2`

C. `3`

D. Output cannot be determined until runtime

E. There is no output due to a syntax error.

QUESTION 25

What is the output by the line marked `//3` ?

A. `12`

B. `6`

C. `4`

D. Output cannot be determined until runtime

E. There is no output due to an error.

```
class A{
  int num;
  public A(int n) {
     num=n;
  }
  public int yeet() {
     return num*2;
  }
}
class B extends A{
  public B(int n) {
     super(n*2);
  }
  public int fun() {
     return yeet()*3;
  }
}
class C extends A{
  public C(int n) {
     super(n*3);
  }
  public int yeet() {
     return super.yeet()*2;
  }
}

///////client code//////////////////
A a=new A(2);
System.out.println(a.yeet()); //1

C c=new C(1);
System.out.println(c); //2

A b=new B(2);
System.out.println(b.fun());  //3
```

QUESTION 26

What is output by the code to the right?

A. `false true`

B. `false false`

C. `true false`

D. `true true`

E. There is no output due to an error

```
out.print
  ("jebron_lam35".matches
              ("\\w+")+" ");
out.print
        ("Getgrinched".matches
              ("[a-z]"));
```

QUESTION 27

What is output by the code to the right?

A. `[0, 1, 2, 3, 4]`

B. `[-3, -2, -1, 3, 1]`

C. `[-2, -1, 2, 1, 2]`

D. `[-2, -1, 2, 2, 1]`

E. `[-1, 1, 1, 2, 3]`

```
int [][] mat = new int[5][5];
for(int i = 0; i < mat.length;i++)
  for(int j = 0; j < mat[i].length;j++) {
    mat[j][i] = i - j;
    mat[i][j] = j - i;
    mat[i][i] = i;
    mat[j][j] = j;
  }
out.println(Arrays.toString(mat[3]));
```

QUESTION 28

What is output by the line marked `//q28`?

A. `35`

B. `32`

C. `45`

D. `13`

E. `47`

QUESTION 29

What is output by the line marked `//q29`?

A. `5`

B. `7`

C. `-2`

D. `4`

E. `2`

QUESTION 30

What is output by the line marked `//q30`?

A. `18`

B. `12`

C. `29`

D. `24`

E. `21`

```
public static int recur(int j) {
 if(j <= 0) {
  return 0;
 }
 if(j >= 50) {
   return 0;
 }
 if(j % 5 == 0) {
  return recur(j-=1) + 3;
 }
 else {
  return recur(j += 2) + 2;
 }
}
///////client code//////////////////
out.println(recur(25));  //q28
out.println(recur(46));  //q29
out.println(recur(34));  //q30
```

QUESTION 31

What is the bit storage of the primitive type `float`?

A. `64` B. `8` C.`4` D. `16` E.`32`

QUESTION 32

What is output by the code to the right?

A. `215`

B. `17`

C. `11`

D. `221`

E. There is no output due to an error.

QUESTION 33

Which of the following data structures is implemented by the `LinkedList` class?

A. Priority Queue

B. Queue

C. Binary Search Tree

D. HashTable

E. Stack

```
LinkedList<Integer>list;
list=new LinkedList<Integer>();
list.add(9);
list.add(8);
list.add(3);
list.add(212);
out.println(list.poll()+list.poll());
```

QUESTION 34

What could replace **<1*>** in the code to the right so that the `resize` works as intended?

A. `new E[len]`
B. `new Object[len]`
C. `(E[])(new Object[len])`
D. More than one of the above.
E. None of the above.

QUESTION 35

Which of the following could replace **<2*>** in the code to the right so that the output by the line marked `//1` is the following?

```
Yeet Grinched Coding null
```

A. `Arrays.toString(list);`
B. `Arrays.toString(list).replaceAll("[\\[\\],]+", "") ;`
C. `Arrays.toString(list).replaceAll("[[],]+", "") ;`
D. `Arrays.toString(list).replaceAll("[\\[\\],]", "") ;`
E. More than one of the above.

QUESTION 36

Assuming that **<1*>** and **<2*>** are filled correctly, what is output by the line marked `//2` in the client code to the right ?

A. 6
B. 8
C. 5
D. 7
E. There is no output due to a syntax error.

QUESTION 37

Assuming that **<1*>** and **<2*>** are filled correctly, what is output by the line marked `//3` in the client code to the right ?

A. `Yeet Grinched Coding Sam Ben null null null`
B. `Yeet Grinched Coding Sam Ben Ben null null`
C. `Yeet Grinched Coding Sam Ben Sam Ben null`
D. `Yeet Grinched Coding Sam Ben`
E. There is no output due to a compile error.

QUESTION 38

What data structure is demonstrated by the `Structure` class ?

A. a linked list
B. a queue
C. a stack
D. a list
E. a set

```
class Structure<E>{

  int size;
  E[] list;

  public Structure() {
    resize(1);
    size=0;
  }
  public void add(E e) {
    if(contains(e))return;
    list[size++]=e;
    if(size==list.length)
      resize(list.length*2);
  }
  public void resize(int len) {
    E[]list2=<1*>;
    if(len!=1)System.arraycopy
      (list,0,list2,0,list.length);
    list=list2;
  }
  public boolean contains(E e) {
    boolean b=false;
    for(E d:list)
      b|=e.equals(d);
    return b;
  }
  public String toString() {
    return <2*>;
  }
}


/////////////client code////////////
Structure<String>s;
s=new Structure<String>();
s.add("Yeet");
s.add("Grinched");
s.add("Coding");
System.out.println(s);     //1

s.add("Sam");
s.add("Ben");
s.add("Sam");
s.add(new String("Ben"));
System.out.println(s.size); //2
System.out.println(s);      //3
```

QUESTION 39

What is output by the code segment to the right?

A. `Nope`

B. `Nope/ by 0`

C. `/ by 0`

D. `4`

E. There is no output due to a syntax error.

```
try {
  int g=256;
  for(int h=8;h>=0;h/=2)
    g/=h;
  System.out.print(g);
}
catch(Exception e) {
  System.out.print("Nope");
}
catch(ArithmeticException a) {
  System.out.print("/ by 0");
}
```

QUESTION 40

What is the value of the following postfix expression (use integer division) :

212 200 - 4 + 4 / 2 / 6 *

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019 Virtual Challenge Meet #3
Computer Science - KEY

1) C
2) C
3) A
4) B
5) E
6) C
7) D
8) C
9) D
10) C

11) C
12) D
13) E
14) B
15) C
16) D
17) B
18) D
19) E
20) B

21) C
22) D
23) A
24) D
25) E
26) C
27) B
28) C
29) D
30) D

31) E
32) B
33) B
34) C
35) E
36) C
37) A
38) E
39) E
40) 12

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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